

Automated Counting of White-Tailed Deer in Vehicle-Mounted Thermal Video: Detector Benchmarking and a Staged Error Decomposition

Taminul Islam^{1*}, Toqi Tahamid Sarker¹, Seth J. Morelock²,
Guillaume Bastille-Rousseau², Khaled R. Ahmed¹

^{1*}School of Computing, Southern Illinois University, Carbondale, IL,
62901, USA.

²Center for Wildlife Sustainability, Southern Illinois University,
Carbondale, IL, 62901, USA.

*Corresponding author(s). E-mail(s): taminul.islam@siu.edu;

Contributing authors: toqitahamid.sarker@siu.edu;

seth.morelock@siu.edu; gbr@siu.edu; khaled.ahmed@siu.edu;

Corresponding author. Taminul Islam, School of Computing, Southern Illinois University, Carbondale, IL 62901, USA. Email: taminul.islam@siu.edu

ORCID. Taminul Islam, 0000-0002-7803-4564; Toqi Tahamid Sarker, 0000-0003-2482-8059; Guillaume Bastille-Rousseau, 0000-0001-6799-639X; Khaled R. Ahmed, 0000-0002-3707-4316.

Declarations

Funding. This research was funded by the Federal Aid Wildlife Restoration Fund, grant number W87R.

Competing interests. The authors declare no competing interests.

Ethics approval. Ethical review and approval were not required for this study because it was purely observational: all footage was recorded from public roads with a vehicle-mounted thermal camera, and no animal was approached, handled, captured or otherwise disturbed.

Consent to participate and for publication. Not applicable.

Data availability. The corpus and the per-individual track annotations will be deposited in Zenodo upon acceptance, and the DOI will be provided at that point.

Code availability. The evaluation harness and the decomposition tooling will be deposited in Zenodo together with the data upon acceptance.

Author contributions. Conceptualization, T.I., G.B.-R. and K.R.A.; methodology, T.I. and K.R.A.; software, T.I. and T.T.S.; validation, T.I.; formal analysis, T.I.; investigation, T.I., S.J.M., G.B.-R. and K.R.A.; resources, S.J.M., G.B.-R. and K.R.A.; data curation, T.I. and S.J.M.; writing, original draft preparation, T.I.; writing, review and editing, T.I., T.T.S., S.J.M., G.B.-R. and K.R.A.; visualization, T.I.; supervision, G.B.-R. and K.R.A.; project administration, G.B.-R. and K.R.A.; funding acquisition, G.B.-R. and K.R.A. All authors have read and agreed to the published version of the manuscript.

Acknowledgements. We thank the Illinois Department of Natural Resources. This work used the Delta and DeltaAI systems at the National Center for Supercomputing Applications.